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**FORM TP 2011057**

MAY/JUNE 2011

**C A R I B B E A N  E X A M I N A T I O N S  C O U N C I L**

**SECONDARY EDUCATION CERTIFICATE EXAMINATION**

# CHEMISTRY

**Paper 02 – General Proficiency**

***2 hours and 30 minutes***

**READ THE FOLLOWING DIRECTIONS CAREFULLY.**

1. **This paper consists of SIX compulsory questions in THREE sections.**
2. **Write your answer to EACH question in the space provided in this answer booklet.**
3. **Where appropriate, ALL WORKING MUST BE SHOWN in this booklet.**
4. **Return this booklet at the end of the examination.**
5. **The use of silent, non-programmable calculators is allowed.**

**DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.**

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## SECTION A

**Answer ALL questions in this section.**

**Write your answers in the spaces provided in this booklet.**

**Do NOT spend more than 30 minutes on Question 1.**

**1.** A group of students conducted an investigation to determine the products of electrolysis when copper(II) sulphate was electrolysed using copper electrodes. The following steps indicate the procedure they used.

- The copper cathode was first washed in distilled water, dried and weighed and then connected to a suitable circuit.
- A current of 0.2 Amps was passed through the solution and the starting time recorded.
- After 5 minutes, the switch was turned off and the cathode was removed, rinsed with distilled water, dried in air and then weighed.
- The cathode was returned to the circuit and the process was repeated, recording the mass of cathode at five-minute intervals. After each interval, the cathode was rinsed in distilled water, dried in air and weighed until a total of five readings was taken.
- Table 1 shows the initial mass of the cathode and its mass for three of the five readings.

**TABLE 1: EXPERIMENTAL RESULTS**

| Time (min) | Mass of Cathode (g) | Actual Mass of Copper Deposited (g) |
|---|---|---|
| 0 | 11.80 | |
| 5 | 11.83 | 0.03 |
| 10 | | 0.05 |
| 15 | 11.87 | |
| 20 | | 0.09 |
| 25 | 11.91 | |

(a) (i) Define the term ‘electrolysis’.

_____

_____

_____

**( 2 marks)**

(ii) State TWO differences between the ‘anode’ and the ‘cathode’.

_____

_____

**( 2 marks)**

(iii) Draw a diagram to show a circuit for carrying out this investigation. Label the cathode and the electrolyte.

**Diagram of Circuit used in the Electrolysis of Copper Sulphate**

**( 3 marks)**

(iv) Complete Table 1 by recording the mass of the cathode after 10 minutes and 20 minutes and the mass of copper deposited after 15 and 25 minutes.

**( 2 marks)**

(v) Plot a graph of actual mass of copper deposited versus time using the axes in Figure 1 on page 5. **( 3 marks)**

(vi) Use the graph to predict the mass of copper that would be deposited after 28 minutes.

_____

**( 1 mark )**

(vii) Write an ionic equation to represent the reaction at the cathode.

_____

**(2 marks)**

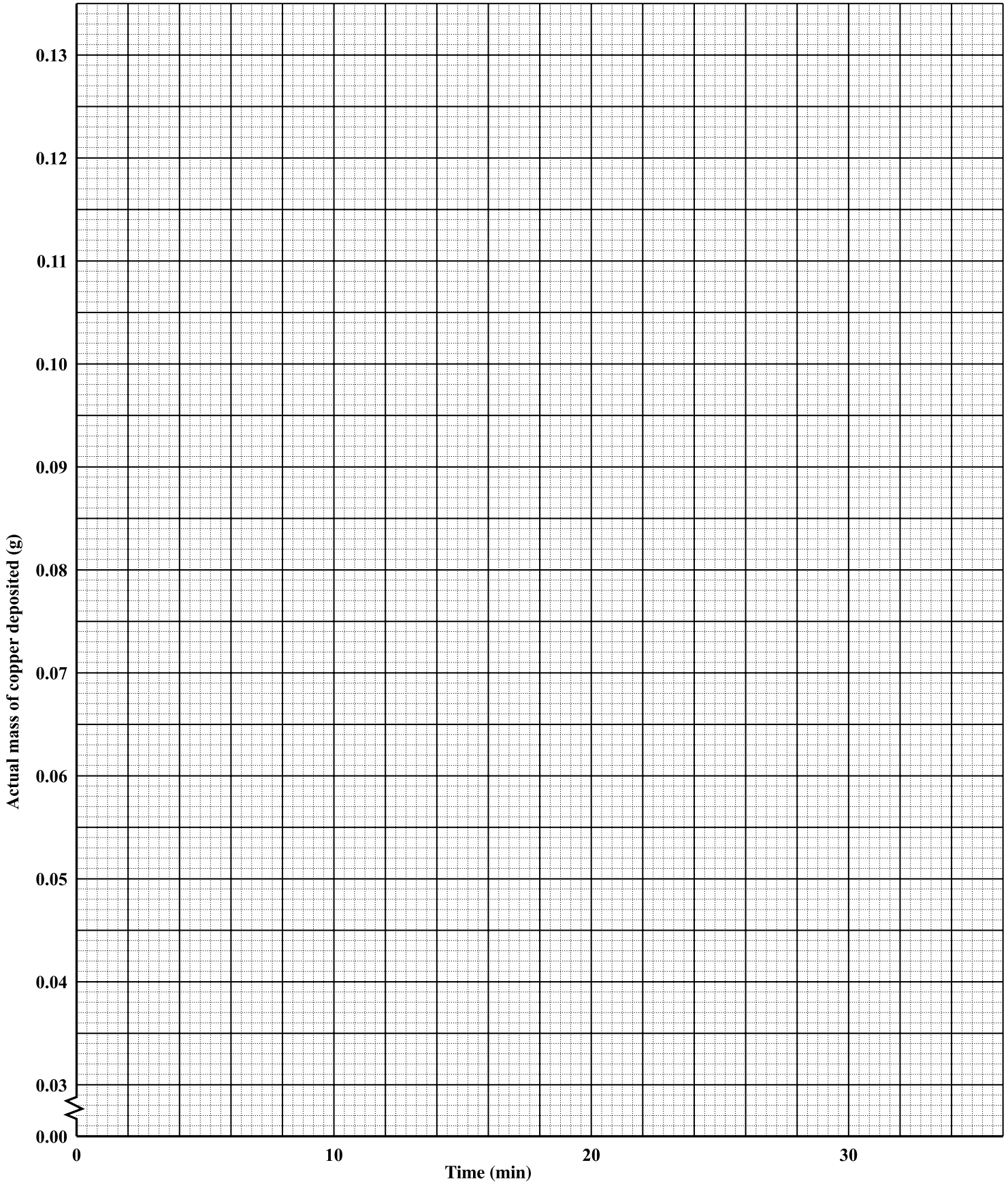



**Figure 1. Actual mass of copper versus time**

(viii) Calculate the quantity of electricity that passed through the copper sulphate solution when 0.2 A of current flowed for 20 minutes.

(Quantity of electricity = current × time; 1 F = 96 500 C)

**( 2 marks)**

(ix) Calculate the number of moles of copper that would be formed.

**( 2 marks)**

(x) Calculate the mass of copper that would be formed.

(Relative atomic mass: Cu = 64; 1 F = 96 500 C)

**( 1 mark )**

(xi) Suggest a reason for the difference between the mass of copper obtained in (x) above and that recorded in Table 1.

_____

_____

**( 1 mark )**

(xii) The circuit used in this experiment was modified by replacing the copper electrodes with graphite electrodes. State ONE difference in the reaction at the anode when graphite electrodes are used.

Difference at the anode

_____

_____

**( 1 mark )**

(b) Complete Table 2 by writing the observations for the tests carried out on Salt S.

**TABLE 2. OBSERVATIONS FOR TESTS ON SALT S**

| Test | Observation | Inference |
|---|---|---|
| A sample of Salt S is heated strongly in a dry test tube. Moist red and blue litmus is held at the mouth of the test tube. | • | An acidic gas is produced. |
| | • | Nitrate ions are present. |
| To an aqueous solution of Salt S, aqueous sodium hydroxide is added gradually until in excess. | | $Zn^{2+}$ are present. |

**( 3 marks)**

**Total 25 marks**

**2.** Depending on their structures and properties, solids may be classified as having metallic, giant covalent, simple molecular, or ionic lattice structures. The structure and conductivity of selected solids are to be summarized in Table 3.

(a) Complete Table 3 to show the structure and conductivity of EACH solid substance listed. The first one is done as an example.

**TABLE 3. STRUCTURE AND CONDUCTIVITY OF SELECTED SOLIDS**

| Solid | Structure | Conducts Electricity in Solid State |
|---|---|---|
| Magnesium chloride | Ionic | No |
| Iodine | | |
| Zinc | | |
| Diamond | | |

**( 6 marks)**

(b) Magnesium chloride and potassium nitrate have ionic lattice structures. Figure 2 on page 9 shows the solubility curves for magnesium chloride and potassium nitrate. Use the information in Figure 2 to answer the questions below.

(i) Which of the two compounds, magnesium chloride or potassium nitrate, has the GREATER solubility at 15 °C?

_____

**( 1 mark )**

(ii) Calculate the mass of potassium nitrate that would be deposited when its saturated solution is cooled from 80 °C to 20 °C.

_____

_____

_____

_____

**( 2 marks)**

(iii) Determine the temperature at which the solubilities of magnesium chloride and potassium nitrate are equal. **( 1 mark )**

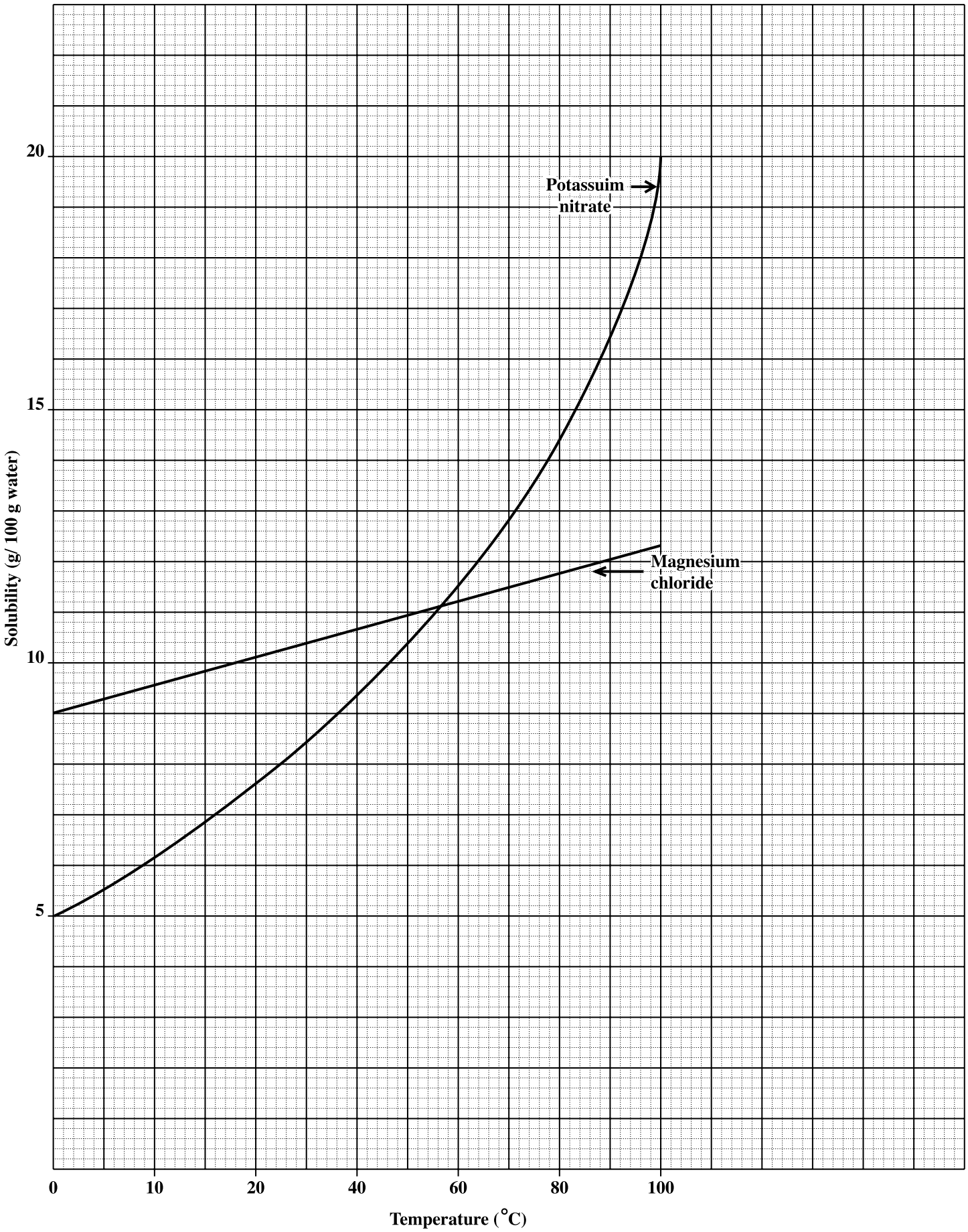



**Figure 2. Solubility curves of magnesium chloride and potassium nitrate**

(c) Draw the bonding diagram to show how magnesium combines with chlorine to form magnesium chloride.

**( 2 marks)**

(d) Based on the use and purpose of solid air fresheners, would you classify them as having simple molecular, giant covalent, or ionic lattices? Explain your answer.

_____

_____

_____

_____

**( 3 marks)**

**Total 15 marks**

**3.** (a) Anaerobic fermentation occurs in the production of wines using yeast.

(i) Define 'anaerobic fermentation'.

Definition: _____

_____

_____

**( 2 marks)**

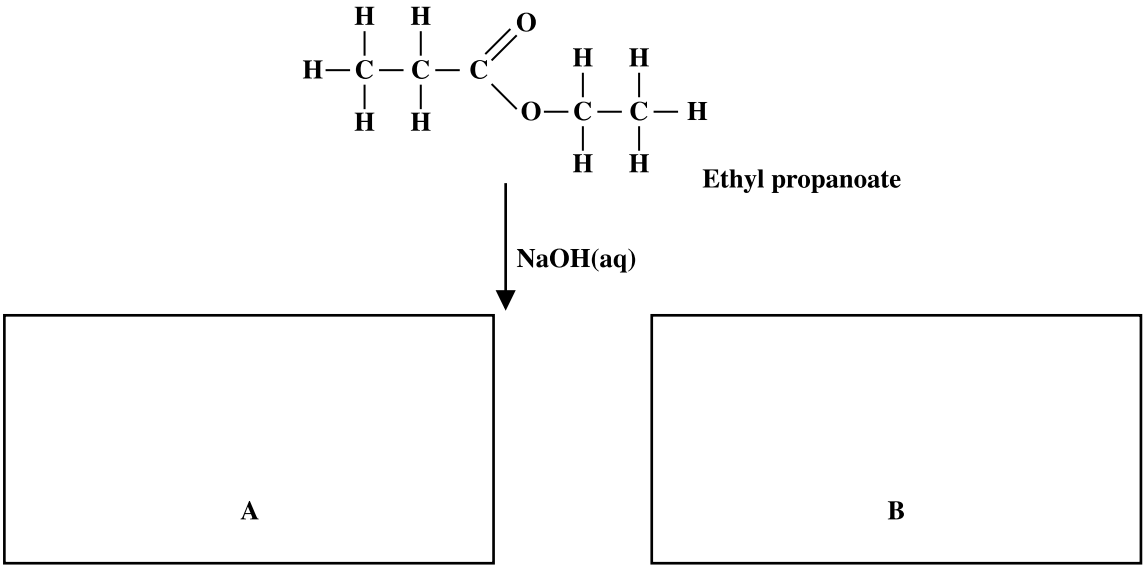
(ii) Give ONE reason why high temperatures are NOT suitable for anaerobic fermentation.

Reason: _____

_____

**( 1 mark )**

(b) Soaps are formed from the alkaline hydrolysis (saponification) of natural fats and oils, which are esters. Ethyl propanoate, an ester of relatively low molecular weight, is hydrolysed by aqueous sodium hydroxide. Draw FULLY displayed structures of the hydrolysis products, **A** and **B**.




**( 4 marks)**

(c) Name ONE by-product of the saponification of fats and oils, and state ONE disadvantage of synthetic cleaning agents over soaps.

By-product: _____

Disadvantage: _____

**( 2 marks)**

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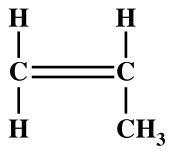
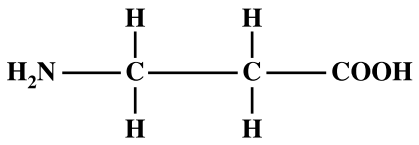
(d) The structures of propene and 3-amino propanoic acid, an amino acid, are shown below:



**Propene**



**3-amino propanoic acid (amino acid)**

(i) Draw the structure of the polymer formed from propene. Use THREE units of the monomer to demonstrate your answer.

**( 2 marks)**

(ii) What type of polymerisation would the 3-amino propanoic acid undergo? State the general name for the type of polymer formed from 3-amino propanoic acid.

Type of polymerisation: _____

**( 1 mark )**

General name: _____

**( 1 mark )**

(iii) Outline a chemical test that can be used to distinguish between propene and its polymer.

_____

_____

**( 2 marks)**

**Total 15 marks**

GO ON TO THE NEXT PAGE

## SECTION B

**Answer ALL questions in this section.**

**Write your answers in the space provided after EACH question in this answer booklet.**

**4.** (a) (i) Potassium carbonate ($K_2CO_3$) and zinc metal (Zn) are both treated with an excess of dilute nitric acid in separate experiments. State the expected observations and write balanced chemical equations to explain any reactions that take place. Give the chemical test(s) which may be used to identify any gases produced. Record your answer in a table using the headings: Test, Observations, Equations. **(10 marks)**

(ii) Outline a method that can be used to obtain a solid sample of a salt where **EITHER** the $K_2CO_3$ **OR** Zn completely reacts with the dilute nitric acid as described in (a) (i) above. **( 2 marks)**

(b) An experiment was performed in three parts (A, B and C) to investigate the factors that are responsible for corrosion (rusting) in iron. Each of three shiny new iron rods was placed in a container and treated as described below, and left for two days.

Part A: The water is boiled and the container is sealed from the atmosphere.

Part B: No water is present and the container is sealed off from the atmosphere.

Part C: Water is present and the container is open to the atmosphere.

Explain the observations expected for EACH part.

**( 3 marks)**

**Total 15 marks**

**Write the answer to Question 4 here.**

**Write the answer to Question 4 here.**

**5.** (a) Some students carried out Experiments I and II below in order to determine the relative reactivity of iron, copper and an unknown Metal X, which has a valency of 2.

Experiment I : A piece of Metal X was placed into a solution of copper(II) sulphate. Metal X was copper plated and the solution gradually became colourless.

Experiment II : A piece of Metal X was placed into a solution of iron(II) sulphate. There was no change to the Metal X or the solution.

(i) Use the information given to place Metal X, copper and iron in the order in which they will appear in the reactivity series, starting with the MOST reactive. **( 2 marks)**

(ii) Write an ionic equation for the reaction between Metal X and copper(II) sulphate. Include the relevant state symbols. **( 2 marks)**

(b) An alloy of copper and aluminium is used in the manufacture of aircraft.

(i) What is an alloy? **( 1 mark )**

(ii) Name the alloy of copper and aluminium. **( 1 mark )**

(iii) State TWO properties of aluminium that make the alloy in (b) (ii) a good choice to be used in the manufacture of aircraft. **( 3 marks)**

(iv) List TWO other uses of this alloy. **( 2 marks)**

(c) Copper is a product of the reaction that occurs when dry ammonia is passed over a sample of heated copper (II) oxide. The equation for the reaction is given below:

$$2NH_3 + 3CuO \rightarrow 3Cu + 3H_2O + N_2 .$$

Calculate the mass of copper produced if 0.12 $dm^3$ of nitrogen is produced at room temperature and pressure (rtp). **( 4 marks)**

(Relative atomic mass of Cu = 64; One mole of gas occupies 24 $dm^3$ at rtp.)

**Total 15 marks**

**Write the answer to Question 5 here.**

**Write the answer to Question 5 here.**

## SECTION C

**Write your answer to this question in the space provided at the end of this booklet.**

**6.** (a) Athletes who use high-protein powders to supplement their diets are advised to mix them with cold or lukewarm water rather than boiling water. Suggest a reason for this. **( 2 marks)**

(b) Meat is an excellent source of protein but it is usually necessary to tenderize meat for human consumption. Both natural and commercial tenderizers can be used to tenderize meat.

(i) Explain how natural OR commercial tenderizers work when used to tenderize meat. **( 3 marks)**

(ii) List ONE advantage and ONE disadvantage of using commercial tenderizers over natural tenderizers. **( 2 marks)**

(iii) Baking soda, $NaHCO_3$, can also be used to tenderize meat. Suggest why this is possible. **( 2 marks)**

(c) The vitamin C content in orange juice can be determined by titrating it with iodine solution using starch as the indicator. Vitamin C reacts with the iodine to form iodide ions. When all the vitamin C is used up the excess iodine reacts with the starch indicator.

A sample of orange juice was divided into two equal portions, A and B. Sample B was boiled then left to cool. Both samples were then titrated against iodine solution.

(i) What colour change would you expect at the end point in BOTH cases? **( 1 mark )**

(ii) Which sample, A or B, would require the LARGER volume of iodine? Explain your answer. **( 3 marks)**

(iii) With reference to the oxidation number of iodine, explain whether vitamin C is acting as an oxidizing or a reducing agent when it reacts with iodine. **( 2 marks)**

**Total 15 marks**

**Write your answer to Question 6 here.**

**Write your answer to Question 6 here.**

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**END OF TEST**